

Euclidean group sparsity does not force an N -term vector-valued RKBS representer

New partial obstruction for arXiv:2509.26371

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Status. New partial result; the source’s structural characterization remains open. For every $N \geq 2$, we construct a finite integral vector-valued RKBS with output space $\mathbb{R}^2$ and a regularized least-squares problem on N data sites whose unique minimizer requires exactly $2N - 1$ kernel sites. Its RKBS norm is the standard group- ℓ_2 penalty. Thus this basic form of groupwise sparsity alone cannot improve the source’s Nd bound to N ; when $d = 2$, the example misses the $2N$ bound by only one.

Source question

Dummer, Heeringa, and Iglesias [1] prove representer theorems for integral and neural vv-RKBSs which, for N measurements in $\mathbb{R}^d$, give at most Nd terms. In their future-work section they ask what additional structure ensures at most N terms, and suggest that groupwise sparsity could play a central role.

are always mapped to finite-dimensional measurements for fitting.

For optimization problems over $f \in \mathcal{B}$ with such data $\{(x_n, y_n)\}_{n=1}^N \subseteq X \times \mathbb{R}^d$, the representer theorem then only guarantees representations of the form

$$f^\dagger(x) = \sum_{m=1}^{Nd} K_{\mathcal{U}}(\cdot, w_m) u_m \quad (7.4)$$

Hence, the number of terms is not necessarily bounded by N . As argued in this paper, we postulate that this limitation arises because the regularizers used so far do not promote groupwise sparsity, unlike, for instance, the structured sparsity regularizers studied by Bach et al. [3]. Achieving such improvements would likely require a more explicit use of the vv-RKBS structure rather than general convexity arguments in topological vector spaces as in Bredies and Carioni [12], which in turn must rely on the use of finite-dimensional measurements.

Moreover, our representer theorems are tailored to integral and neural vv-RKBSs. Similar to the work of Wang et al. [50], which investigates when scalar RKBSs admit a representer theorem, it would be interesting to study under what conditions our (adjoint pair of) vv-RKBSs admit one. In particular, an important open question is what additional structure is needed to ensure representations with at most N terms in the expansion. We expect that the groupwise sparsity mentioned above could play a central role.

Source evidence: arXiv:2509.26371, p. 33, equation (7.4) and the following open question.

Proof intuition

In a finite discrete integral vv-RKBS, the total-variation norm of a vector measure is exactly a sum of Euclidean coefficient norms. This is ordinary group lasso, one output vector per kernel site. We expose a large flat face of that norm with a dual matrix G . Each of $2N - 1$ atoms lies on the face because $G^T a_j$ is its unit output direction. The atoms’ prediction matrices $a_j s_j^T$ are linearly independent. Consequently equality

in the dual bound first pins every coefficient to its prescribed nonnegative ray and then pins every ray scale to one. A squared loss is centered so that this exposed point satisfies the exact optimality equations.

Finally, we complete the training feature matrix to an invertible square matrix on the whole finite domain. This removes a possible loophole: the same minimizer cannot secretly have a shorter kernel expansion away from the training data.

Theorem 1 (a $2N - 1$ -site group- ℓ_2 obstruction). *For every integer $N \geq 2$, there exist finite sets X, Ω , a scalar feature map $\phi : X \times \Omega \rightarrow \mathbb{R}$, an integral vv-RKBS $\mathcal{B} \subseteq \{f : X \rightarrow \mathbb{R}^2\}$ with kernel sections*

$$K_{\mathcal{U}}(x, w)u = \phi(x, w)u,$$

N distinct training sites $x_1, \dots, x_N$, and targets $t_i \in \mathbb{R}^2$ such that

$$\min_{f \in \mathcal{B}} \left\{ \frac{1}{2N} \sum_{i=1}^N \|f(x_i) - t_i\|_2^2 + \frac{1}{N} \|f\|_{\mathcal{B}} \right\} \quad (1)$$

has a unique minimizer. Every kernel expansion of that minimizer uses exactly $2N - 1$ distinct sites of Ω , and $\|f\|_{\mathcal{B}}$ is the group- ℓ_2 coefficient norm.

Proof. Put $m = 2N - 1$. Let $e_1, \dots, e_N$ be the standard basis of $\mathbb{R}^N$, and let $G \in \mathbb{R}^{N \times 2}$ have first row $(1, 0)$, second row $(0, 1)$, and all other rows zero. We specify m feature columns $a_j \in \mathbb{R}^N$ and unit directions $s_j \in \mathbb{R}^2$. The first three are

j	1	2	3
a_j	e_1	e_2	$(e_1 + e_2)/\sqrt{2}$
s_j	$(1, 0)$	$(0, 1)$	$(1, 1)/\sqrt{2}$

For each $k = 3, \dots, N$, add two more pairs

$$(a_{k,0}, s_{k,0}) = (e_1 + e_k, (1, 0)), \quad (a_{k,1}, s_{k,1}) = (e_2 + e_k, (0, 1)).$$

There are $3 + 2(N - 2) = m$ pairs, and every one satisfies

$$G^T a_j = s_j, \quad \|s_j\|_2 = 1. \quad (2)$$

The matrices $b_j = a_j s_j^T \in \mathbb{R}^{N \times 2}$ are linearly independent. Indeed, for each $k \geq 3$, row k of a linear relation separately forces the coefficients of $b_{k,0}$ and $b_{k,1}$ to vanish. Among the first three matrices, the off-diagonal entries force the coefficient of b_3 to vanish, and the two diagonal entries then dispose of b_1, b_2 . Also the matrix $A = [a_1 \cdots a_m] \in \mathbb{R}^{N \times m}$ has rank N : besides e_1, e_2 , its column span contains $a_{k,0} - a_1 = e_k$.

Choose an $(m - N) \times m$ matrix C so that $F = \begin{bmatrix} A \\ C \end{bmatrix}$ is invertible. Take $X = \{x_1, \dots, x_m\}$, $\Omega = \{w_1, \dots, w_m\}$, and $\phi(x_i, w_j) = F_{ij}$. These finite sets are discrete and compact, so all features are continuous. Define

$$f_c(x) = \sum_{j=1}^m \phi(x, w_j) c_j, \quad \|f_c\|_{\mathcal{B}} = \sum_{j=1}^m \|c_j\|_2.$$

This is the finite discrete integral vv-RKBS obtained from vector measures on Ω . Invertibility of F makes the coefficient tuple $c = (c_1, \dots, c_m)$ unique, so the displayed formula really is its RKBS norm.

Let the candidate coefficients be $c_j^0 = s_j$. Its training prediction matrix, with one output row per training point, is

$$Y_0 = A c^0 = \sum_{j=1}^m a_j s_j^T = \sum_{j=1}^m b_j.$$

Set the target matrix to $T = Y_0 + G$. In coefficient form, (1) is the convex objective

$$J(c) = \frac{1}{2N} \|Ac - T\|_F^2 + \frac{1}{N} \sum_{j=1}^m \|c_j\|_2. \quad (3)$$

At c^0 , the residual is $-G$. Since no s_j is zero, the rowwise optimality equation is, by (2),

$$\frac{1}{N} a_j^T (-G) + \frac{1}{N} s_j = 0.$$

Hence c^0 minimizes J .

It remains to prove uniqueness and the support claim. Strict convexity of the squared loss in the prediction Ac , together with convexity of the norm, implies that every minimizer has prediction Y_0 . For any c satisfying $Ac = Y_0$, the same matrix G is a dual certificate:

$$\begin{aligned} \sum_{j=1}^m \|c_j\|_2 &\geq \sum_{j=1}^m \langle s_j, c_j \rangle = \langle G, Ac \rangle_F = \langle G, Y_0 \rangle_F \\ &= \sum_{j=1}^m \langle G^T a_j, s_j \rangle = m. \end{aligned}$$

The candidate attains equality. Equality term by term in the first inequality forces $c_j = r_j s_j$ with $r_j \geq 0$. The prediction identity then becomes

$$\sum_{j=1}^m r_j b_j = \sum_{j=1}^m b_j.$$

Linear independence of the b_j 's forces every $r_j = 1$. Thus c^0 is the unique minimizer and all its $m = 2N - 1$ coefficients are nonzero. Since F is invertible, every expansion of the resulting function has this same coefficient tuple (after combining repeated sites). No expansion on fewer distinct kernel sites exists. $\square$

What this answers—and what it does not

Taking the measurement operator to be the identity on $\mathcal{U} = \mathbb{R}^2$ gives $d = 2$ in the source's setup. Theorem 1 therefore supplies, for every sample size, an example requiring $2N - 1 > N$ terms. It disproves the sufficiency of the most direct interpretation of the suggested route: the standard Euclidean group- ℓ_2 norm by itself. It also nearly saturates the general $Nd = 2N$ guarantee.

The theorem does *not* say that stronger structured group penalties are ineffective, and it does not characterize the additional geometry needed for an N -term theorem. Those parts of the source question remain open. Three focused upgrade attempts progressed from a three-site $N = 2$ example, to an $N + 1$ -site polygon family, and finally to the $2N - 1$ -site exposed face above. The last construction reaches the dimension of the natural rank-two exposed-face span; no credible route to $2N$ under the same smooth Euclidean certificate remained.

The bundled script independently checks ranks, the exposing identities, the optimality equations, the invertible completion, the equality-case linear system, and random perturbations for $2 \leq N \leq 12$. These are sanity checks only; the general result is proved symbolically above. Exact-title, exact-question, vv-RKBS representer, group-lasso active-set, and $2N - 1$ -support searches through 13 August 2026 found no later source with this construction or an answer to the stated structural question. This was a bounded search, not a systematic bibliographic review.

Reference

- [1] S. Dummer, T. J. Heeringa, and J. A. Iglesias, *Vector-Valued Reproducing Kernel Banach Spaces for Neural Networks and Operators*, arXiv:2509.26371, v3 (2026), especially Section 7.